

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

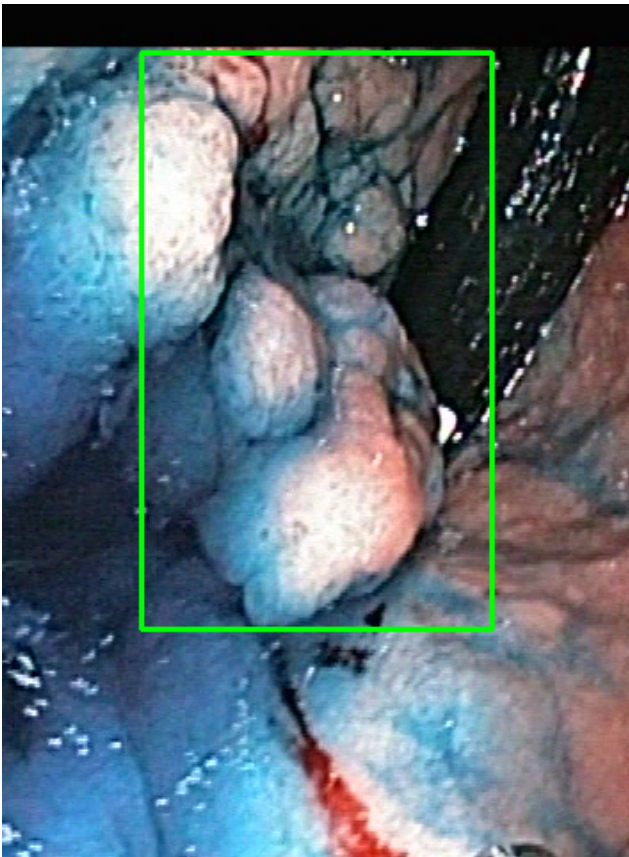
Video ID:	b91de20f-e2ce-45d9-9d2f-df8df1e63c73
Total Frames:	1653
Frame Rate:	25.0 fps (assumed)
Duration:	66.1 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	1530	✓
RT-DETR Detections	2117	✓
MedSAM2 Detections	2365	✓
Consensus (3-Model Agreement)	3	✓
Risk Classification	Count	Percentage
HIGH RISK	1	33.3%
MEDIUM RISK	0	0.0%
LOW RISK	2	66.7%
TOTAL ANALYZED	3	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 83 frames)



Feature	Value	Interpretation
Frame Index	1576	Frame 1576 of 1653
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 72.8%
Redness Score	0.028	Color-based vascularization
Relative Radius	18.9%	Polyp size as % of frame diagonal
Texture Score	0.221	Surface morphology
Vessel Visibility	0.042	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	80.4%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — LIFTED_POLYP [LOW_RISK] (Tracked across 908 frames)



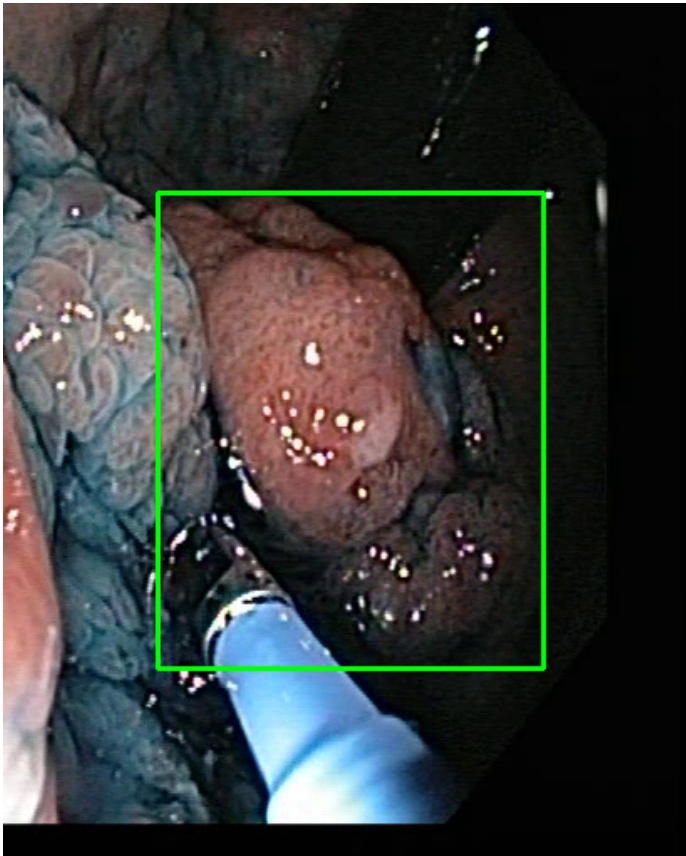
Feature	Value	Interpretation
Frame Index	1196	Frame 1196 of 1653
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 76.6%
Redness Score	0.039	Color-based vascularization
Relative Radius	30.9%	Polyp size as % of frame diagonal
Texture Score	0.407	Surface morphology
Vessel Visibility	0.026	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	84.5%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #3 — MALIGNANT_POLYP [HIGH_RISK] (Tracked across 67 frames)



Feature	Value	Interpretation
Frame Index	987	Frame 987 of 1653
Detection Method	Detector Consensus (YOLO / RT-DETR / MedSAM2)	Detector Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	MALIGNANT_POLYP	Type confidence: 92.0%
Redness Score	0.090	Color-based vascularization

Relative Radius	57.8%	Polyp size as % of frame diagonal
Texture Score	0.744	Surface morphology
Vessel Visibility	0.508	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	80.7%	YOLO / RT-DETR / track confidence

Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.

Low-confidence HIGH RISK detection - Further evaluation recommended

CLINICAL IMPRESSION

Summary:

A total of 3 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 1 (33.3%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 2 (66.7%)

Recommendation:

The presence of HIGH RISK polyps warrants close clinical attention. Consider endoscopic resection or follow-up surveillance depending on clinical context.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.